Finance Research Letters 82 (2025) 107557

Contents lists available at ScienceDirect

Finance Research Letters

journal homepage: www.elsevier.com/locate/frl

Energy, metals, cereals and G7 indices: Russia–Ukraine conflict and risk spillovers

Maria Leone [a,*], Alberto Manelli [a], Roberta Pace [b]

[a] Department of Management, Polytechnic University of Marche, Piazzale Martelli 8, 60121 Ancona, Italy
[b] Department of Industrial Engineering and Economics, University of L'Aquila, Via G. Mezzanotte, 67100 L'Aquila, Italy

ARTICLE INFO

JEL classification:
G15
G01
C32

Keywords:
Geopolitics
G7 financial markets
Commodity markets
Volatility connectedness
Spillovers

ABSTRACT

The economies of each State are increasingly interconnected and depend on international trade. The intricate set of connections and transactions was put to the test during the Russia–Ukraine conflict. The TVP-VAR model is used to investigate the connectedness among G7 stock indices and commodity markets. Results show that spillovers are dynamic and crisis sensitive and the response at the war has been instantaneous and in counter trend. Therefore, the war significantly affected most of the G7 stock prices through commodity prices. This dependence on raw materials makes the G7 countries closely tied to the belligerents more sensitive than others to international crises and conflicts.

1. Introduction

On February 24, 2022 the diplomatic-military clash between Russia and Ukraine, which began eight years earlier in February 2014, worsened and resulted in an invasion of some Ukrainian territories. The war in Ukraine has caused alarming effects on the world economy: the two countries are among major exporter of a wide range of commodities such as crude oil, natural gas, metals and wheat. The exports of the two countries, and of Russia through Ukraine, are considerable in quantity, in terms of their economic amount and of the number of destination countries. Then, the war between Russia and Ukraine has disrupted international trade and commodity markets, and the consequences of the invasion were not contained regionally but rapidly created a significant disruption of the global economy. From these premises it is evident that the war has increased commodity price fluctuations, and markets' contagion (Yousaf et al. (2023)). These effects have been more visible given the increasing level of global trade integration and connectedness among markets that has strengthened economic interdependence. The Russia–Ukraine war together with Covid-19 and the 2008 financial crisis has caused immeasurable risks for the global economy (Zhang and Sun (2023) and Gaio et al. (2022)) and has made it more difficult for stock markets to recover from the effects of Covid-19 (Clancey-Shang and Fu (2023)). The geopolitical unrest that emerged with the increase in commodity prices has amplified the negative effects of the pandemic. In this field Burns and Prager (2024) show that stock market shocks affect commodity swap markets. They indicate that stock market volatility following the pandemic affected the positions of agricultural swap traders. Inacio et al. (2023) suggest a greater impact of the conflict on European diesel prices compared to US because the former market shows a significant difference between the pre and war periods. Lei et al. (2023) consider the volatility of Karachi Stock Exchange and crude oil during crisis periods and demonstrate how after the pandemic and conflict there has been a transmission of shocks and volatility from oil to stock markets. Numerous studies have analyzed the Russia–Ukraine conflict considering the impact that news had on financial markets (Dragomirescu-Gaina

* Corresponding author.
E-mail address: m.leone@univpm.it (M. Leone).

https://doi.org/10.1016/j.frl.2025.107557
Received 27 December 2024; Received in revised form 30 April 2025; Accepted 8 May 2025
Available online 24 May 2025
1544-6123/© 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

et al. (2024) and Ma et al. (2024)) and the economic consequences of geopolitical risk. The results have shown how the latter has had significant impacts on business and stock markets with particular emphasis on volatility (Pan and Sun (2023)), and returns (Lee et al. (2024)).

Although the current literature provides a diverse perspective on the war's initial impact, there is still room for improvement. First, many studies rely on a short sample period spanning approximately one year or less and thus only capture the conflict's early effects or focusing primarily on the relation among stock markets (Cheng et al. (2024)) or between these and oil price (Cui et al. (2024)). In a context such as Russia–Ukraine war, in which commodity prices have shown anomalies and violent fluctuations (Wu et al. (2023)), the aim of the paper is to study the dynamic connectedness among financial and commodity markets in order to achieve the time-varying power and direction of information transfer within a common framework. This is even more important given that the historic low correlation between these two markets has increased as never before during conflict (Umar et al. (2022)). We choose G7 countries because in the financial system these States occupy a crucial position. G7 is an organization of developed countries, with the world's largest industrialized economies and their financial markets have wide global influence. Given that Russia and Ukraine are significant exporters of several commodities, and that the G7 countries — mainly European ones — have traditionally heavily relied on Russia for imports of hard and soft commodities, a more in-depth analysis including a wide range of products (minerals, metals, agricultural) is essential to provide an assessment of the likely consequences of the war on stock markets. So, G7 indices offer a comprehensive picture of how Russia–Ukraine war has changed the relationship between commodity and stock markets.

The reminder of this paper is organized as follows. Section 2 presents the methodology; Section 3 outlines the data used and shows empirical findings; Section 4 offers study's conclusions and implications.

2. Methodology

In order to accurately examine the transmission mechanism of energy markets, a dynamic connectedness TVP-VAR model (Chatziantoniou and Gabauer (2021)) is chosen. This model roots dynamic connectedness measures (Diebold and Yilmaz (2014)) on the results of a TVP-VAR with time-varying covariances (Koop and Korobilis (2014)). This methodology allows not to lose observations and to choose the window size in a non-arbitrary way. This model is also particular suitable for the analysis of financial markets that exhibit episodes, albeit occasional, in which price volatility increases considerably. In these cases, TVP-VAR model outperforms VAR models, sensitive to extreme price variations. The TVP-VAR model is as follows:

$$\begin{aligned} x_t &= \Phi_t x_{t-1} + \varepsilon_t \\ \varepsilon_t | \Omega_{t-1} &\sim N(0, \Sigma_t) \end{aligned} \tag{1}$$

$$\begin{aligned} vec\Phi_t &= vec(\Phi_{t-1}) + \xi_t \\ \xi_t | \Omega_{t-1} &\sim N(0, \Xi_t) \end{aligned} \tag{2}$$

where, Ω_{t-1} illustrates all available information until time $t-1$, x_t and x_{t-1} are $m \times 1$ dimensional vectors, Φ_i and Σ_t are $N \times N$ coefficient matrices, and ε_t, is the vector of the error terms.

To calculate the generalized forecast error variance decomposition (GFEVD) (Koop et al. (1996)) the time-varying parameters model is transformed, using the Wold representation theorem, in a TVP-VMA which is represented as follows:

$$x_t = \sum_{i=1}^{p} \Phi_{it} x_{t-i} + \varepsilon_t = \sum_{j=1}^{\infty} \Lambda_{jt} \varepsilon_{t-j} + \varepsilon_t \tag{3}$$

Since the aim of the analysis is to examine the dynamic connectedness among stock and commodity markets, it is possible to determine different connectedness indices. The total connectedness index (TCI) measures the total information spillover in a system and expresses the average amount of the share of variance of the forecast error of a variable explained by all other variables. The equation of TCI is:

$$C_t^g(K) = \frac{\sum_{i,j=1, i \neq j}^{m} \tilde{\Psi}_{ij,t}^g(K)}{m} \tag{4}$$

The total directional connectedness TO others describes how much a shock in variable i is able to influence all other variables j. The equation of TO is:

$$C_{i \to j,t}^g(K) = \frac{\sum_{j=1, i \neq j}^{m} \tilde{\Psi}_{ji,t}^g(K)}{\sum_{j=1}^{m} \tilde{\Psi}_{ji,t}^g(K)} \tag{5}$$

Similarly, the total directional connectedness FROM others describes how much variable i receives from shocks in all other variables j. The equation of FROM is:

$$C_{i \leftarrow i,t}^g(K) = \frac{\sum_{j=1, i \neq j}^{m} \tilde{\Psi}_{ij,t}^g(K)}{\sum_{j=1}^{m} \tilde{\Psi}_{ij,t}^g(K)} \tag{6}$$

The net total directional connectedness (NET) measures the difference between the total directional connectedness TO others and the total directional connectedness FROM others. It indicates the net contribution of the variable i to the system. The equation of NET is:

$$C_{i,t}^g = C_{i \to j,t} - C_{i \leftarrow i,t}^g \tag{7}$$

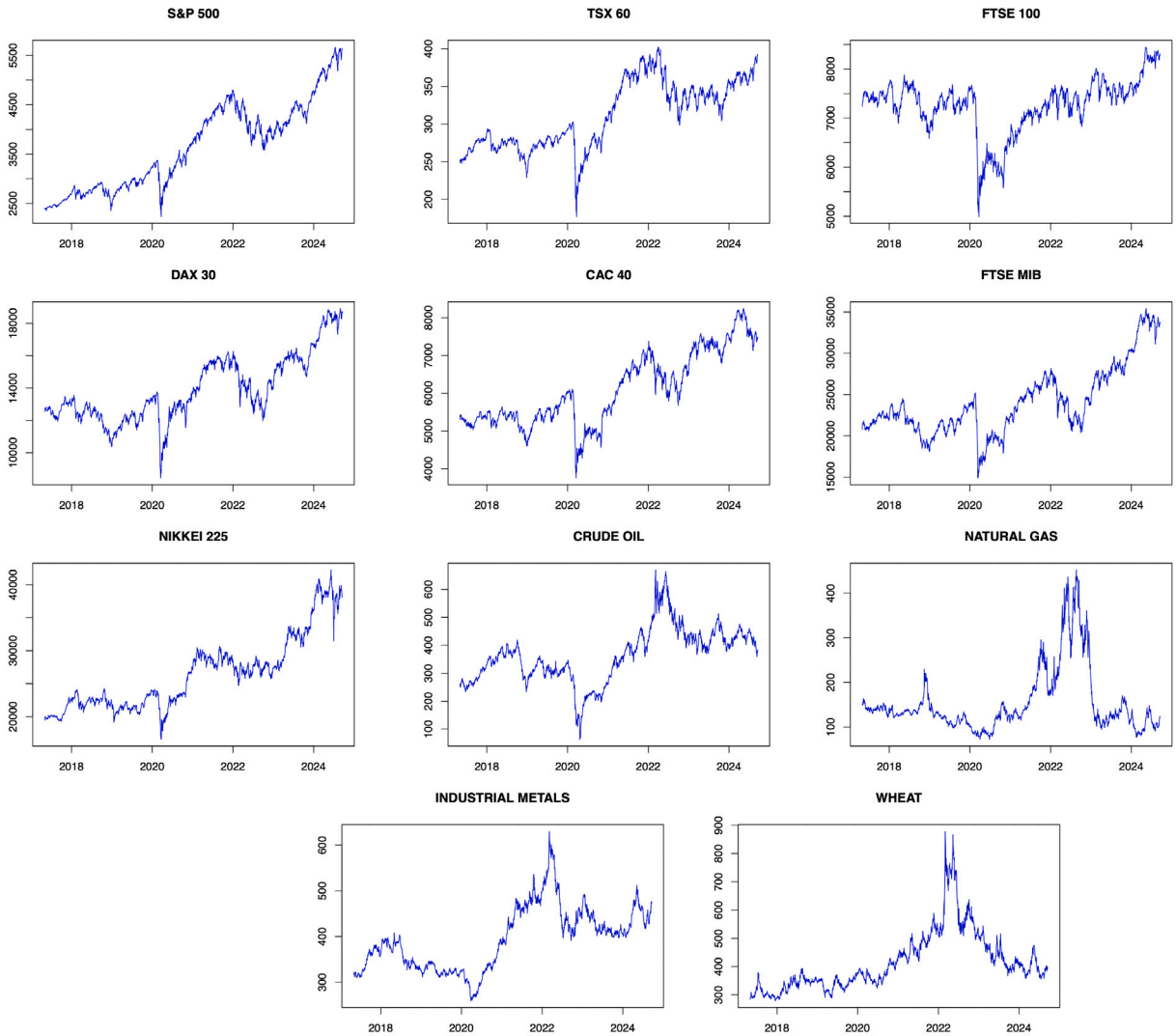

Fig. 1. Daily closing prices of stock indices and commodities.
The above figures represent the daily closing values of G7 stock indices and S&P GSCI Crude oil index, S&P GSCI Natural gas index, S&P GSCI Industrial metals index and S&P GSCI Wheat index, which provide a reliable benchmark for investment performance in the crude oil, natural gas, industrial metals and wheat markets, respectively.

When $C^g_{i,t} > 0$ the variable i is defined a net transmitter because it is influencing all other variables more than being influenced by them. Otherwise, when $C^g_{i,t} < 0$ the variable i is defined a net receiver.

3. Empirical results

In the analysis we examine the prices of the main commodities exported by Russia and Ukraine and the stock market indices of the G7 countries. In particular, our dataset consists of daily returns of S&P 500 (US), TSX 60 (Canada), FTSE 100 (UK), DAX 30 (Germany), CAC 40 (France), FTSE MIB (Italy) and NIKKEI 225 (Japan). Among commodities we consider Crude oil, Natural gas, Industrial metals, and Wheat. We use each market's closing prices from 3 May 2017 to 17 September 2024. The daily closing price returns are defined as follows:

$$y_{i,t} = ln(p_{i,t}) - ln(p_{i,t-1}) \tag{8}$$

where $p_{i,t}$ is the daily price of the each index i at time t.

Fig. 1 illustrates the time plots of the closing prices of stock market indices and commodities in the analyzed period. After the conflict, the indices fell by more than 10%, reaching their minimum between September and October 2022. Conversely, all commodities continued their growth path reaching maximum values in the days immediately following the conflict with an increase

Table 1
Descriptive statistics.

	S&P500	CAC40	DAX30	FTSE100	FTSEMIB	NIKKEI225	TSX60	Crude oil	Natural gas	Industrial Metals	Wheat
Pre-conflict period											
Mean	0.000477	0.000205	0.000131	0.0000281	0.000188	0.000238	0.000322	0.000530	0.000298	0.000421	−0.000095
Median	0.000950	0.000805	0.000676	0.000565	0.00108	0.000558	0.000701	0.002547	0.000389	0.000997	−0.000532
Minimum	−0.127652	−0.130983	−0.130548	−0.115100	−0.185411	−0.062735	−0.142523	−0.568609	−0.191829	−0.041141	−0.797588
Maximum	0.089683	0.080560	0.1041429	0.086670	0.085495	0.077313	0.120824	0.320061	0.198066	0.037335	0.061740
Std. Dev.	0.012634	0.012162	0.012527	0.010857	0.013771	0.011975	0.012600	0.036018	0.031285	0.010308	0.028811
Skewness	−1,0786887	−1,3908662	−1,005327	−1,254439	−2,606204	−0,165413	−1,942948	−3,481586	0,152714	−0,223230	−17,688686
Kurtosis	19,800764	18,079030	17,031155	17,937187	34,138188	4,602859	37,151085	69,643877	4,931768	1,102903	491,604415
ERS	−14.849	−13.600	−13.198	−12.717	−12.900	−9.510	−11.872	−7.203	−14.190	−4.211	−5.950
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
JB	20137.377***	16757.107***	14800.054***	15873.179***	59745.853***	1136.702***	67618.826***	235394.390***	1209.914***	62.308***	12 380 500.503***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Conflict period											
Mean	0.000450	0.000155	0.000386	0.000160	0.000412	0.000616	0.000089	−0.000430	−0.000882	−0.000241	−0.000605
Median	0.000324	0.000462	0.000788	0.000655	0.001205	0.001193	0.000677	0.000546	−0.000495	−0.000503	−0.001976
Minimum	−0.044199	−0.050928	−0.045083	−0.039554	−0.064384	−0.132340	−0.386700	−0.136448	−0.182683	−0.399970	−0.100158
Maximum	0.053952	0.068827	0.076231	0.038391	0.067143	0.097365	0.429100	0.082623	0.133516	0.051553	0.082831
Std. Dev.	0.011302	0.010840	0.010926	0.008286	0.012222	0.013443	0.010777	0.024210	0.042912	0.012847	0.023640
Skewness	−0,213895	0,095715	0,164184	−0,443935	−0,514601	−1,155454	−0,123633	−0,443438	−0,309095	0,149999	0,232290
Kurtosis	1,815172	4,038147	4,966490	3,586071	3,852535	19,125727	0,981081	2,111078	0,814500	0,819535	1,378965
ERS	−7.028	−3.528	−5.186	−5.880	−3.917	−3.107	−4.405	−12.488	−6.580	4.961	−11.472
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
JB	80.518***	402.783***	601.374***	359.721***	385.873***	8688.744***	27.122***	140.525***	26.952***	19.916***55.815***	
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
All sample											
Mean	0.000468	0.000188	0.00022	0.0000744	0.0002662	0.0003706	0.0002412	0.000196	−0.0001131	0.0001995	−0.0002731
Median	0.0008396	0.0007513	0.0006863	0.0005982	0.0011244	0.0007652	0.0006815	0.0021005	0.0001078	0.0005731	−0.0009277
Minimum	−0.1276521	−0.1309835	−0.1305486	−0.1151243	−0.1854114	−0.1323408	−0.1425236	−0.5686094	−0.191829	−0.0411415	−0.7975886
Maximum	0.0896832	0.0805608	0.1041429	0.0866681	0.0854946	0.0973656	0.1208241	0.3200619	0.1980662	0.0515532	0.0828316
Std. Dev.	0.0121839	0.0117157	0.0119917	0.010035	0.0132495	0.0125041	0.0119944	0.0323931	0.0357609	0.011259	0.0271166
Skewness	−0.8425014	−0.9863691	−0.7070767	−1.126306	−2.049338	−0.5893168	−1.494488	−3.168049	−0.1343686	0.0518648	−13.76034
Kurtosis	18.33531	17.65769	17.4005	19.69095	29.87149	14.32415	32.58188	72.44755	5.775569	4.10039	409.2836
ERS	−18.774	−17.413	−16.878	−15.766	−16.794	−12.113	−15.906	−8.590	−17.601	−5.563	−7.127
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
JB	18 188.008***	16 715.340***	15 999.690***	21 676.520***	56 462.447***	9905.544**	67 553.969***	371 622.510***	594.216***	93.352***	12 671 700.699***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Notes: This table presents the descriptive statistics and stationarity properties of the sample.
The sample include the returns of G7 stock indices — US (S&P 500), France (CAC 40), Germany (DAX 30), UK (FTSE 100), Italy (FTSE MIB), Japan (NIKKEI 225) and Canada (TSX 60) – and S&P GSCI Crude oil index, S&P GSCI Natural gas index, S&P GSCI Industrial metals index and S&P GSCI Wheat index.
The sample period — between 2 May 2017 to 17 September 2024 — consists of 1834 observations.
The Pre-conflict period — between 2 May 2017 to 23 February 2022 — consists of 1179 observations
The Conflict period — between 24 February 2022 to 17 September 2024 — consists of 655 observations.
The robustness standard errors*, **, *** represent statistical significance at 10%, 5%, and 1% levels, respectively.

of over 100% compared to the previous year and although showing a decline at the end of 2022, they continue to denote a very high value. In more detail among these commodities, natural gas and wheat are those that recorded the greatest price peaks: between the end of 2021 and April 2022 natural gas increased by more than 200% and wheat by more than 110%. For these reasons, causality is examined by distinguishing two distinct periods: pre-conflict period — spanning from 2 May 2017 to 23 February 2022, a fairly long time interval which includes notably and noteworthy crises — and conflict period — spanning from 24 February 2022 (when Russia invaded Ukraine) to 17 September 2024.

Table 1 presents the descriptive statistics for stock markets and commodities. If we examine the difference between max and min, we note that the greatest gap is recorded in commodity markets. But all series show a positive and close to zero mean except wheat in all periods, and all commodity indices in conflict period. This confirms the growing trend in commodity prices during conflict. The skewness shows negative and positive values and all values, except wheat in pre-conflict period and in all sample, are close to zero, indicating a greater level of skewness. Also, for kurtosis all indices show positive values, and they are higher than 3, indicating a leptokurtic distribution. The results of ERS test confirm that all returns are stationary for all the periods considered and it indicates that the series are DS processes (Elliott et al. (1996) and Laurent and Shi (2022)).

We start our discussion by analyzing the total dynamic connectedness (TCI) and net return (NET) between G7 stock indices and commodity markets. Table 2 shows that the main shock transmitters are the S&P 500, CAC 40, DAX 30, FTSE 100 indices both in pre-conflict and in conflict periods, while FTSE MIB only in conflict period. Conversely, the net shock recipients are NIKKEI 225, TSX 60 and all commodity markets. This is confirmed by the pairwise net directional transmission dominance value which for these last markets shows negative values in all periods analyzed. But there are notable differences among different periods. In fact, TCI — that explains that on average the co-movement of markets, and therefore the risk equality of the entire network — is greater in conflict period (52.77%) than in pre-conflict period (44.98%). In turn, this means that on average 40.89% in pre-conflict and 47.97% in conflict period of the variance of the forecast error of a market return can be explained by the influence of the returns of all other markets. This is also confirmed by Fig. 2, which provides the interconnectedness of the network over time. It shows how the total dynamic return connectedness is variable over time, but always above 25%, indicating, thus, a solid interconnectedness between stock and commodity markets. Nevertheless, this hard connectedness among financial and commodity markets reaches values greater than 50% in periods of tension, such as pandemic and Russia–Ukraine war. In particular, after the pack reached at the beginning of the conflict, the TCI values continued to be high and above 40% until the end of 2024. This very strong connection denotes how in times of significant crisis, extreme market conditions and uncertainty the interconnectedness between stock and commodity markets is significantly consolidated (Lin et al. (2024)). This may be because in the presence of unlooked-for and uncertain events investors are more cautions in diversifying their portfolios and less willing to take risk, this causes a greater interconnectedness between markets. (Adekoya et al. (2021)).

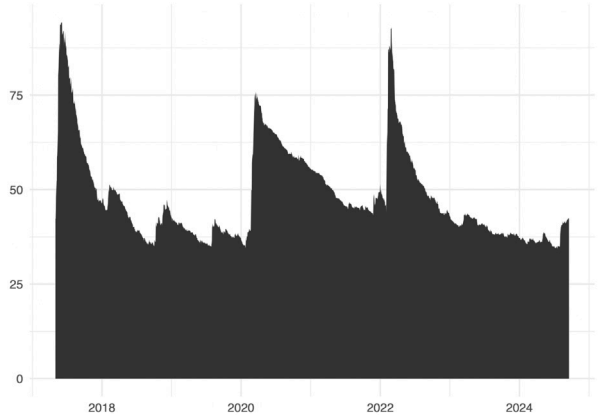

Fig. 2. Dynamic total return spillover.
Note: Results are based on TVP-VAR model with lag length of BIC, and a 10-step-ahead forecast.

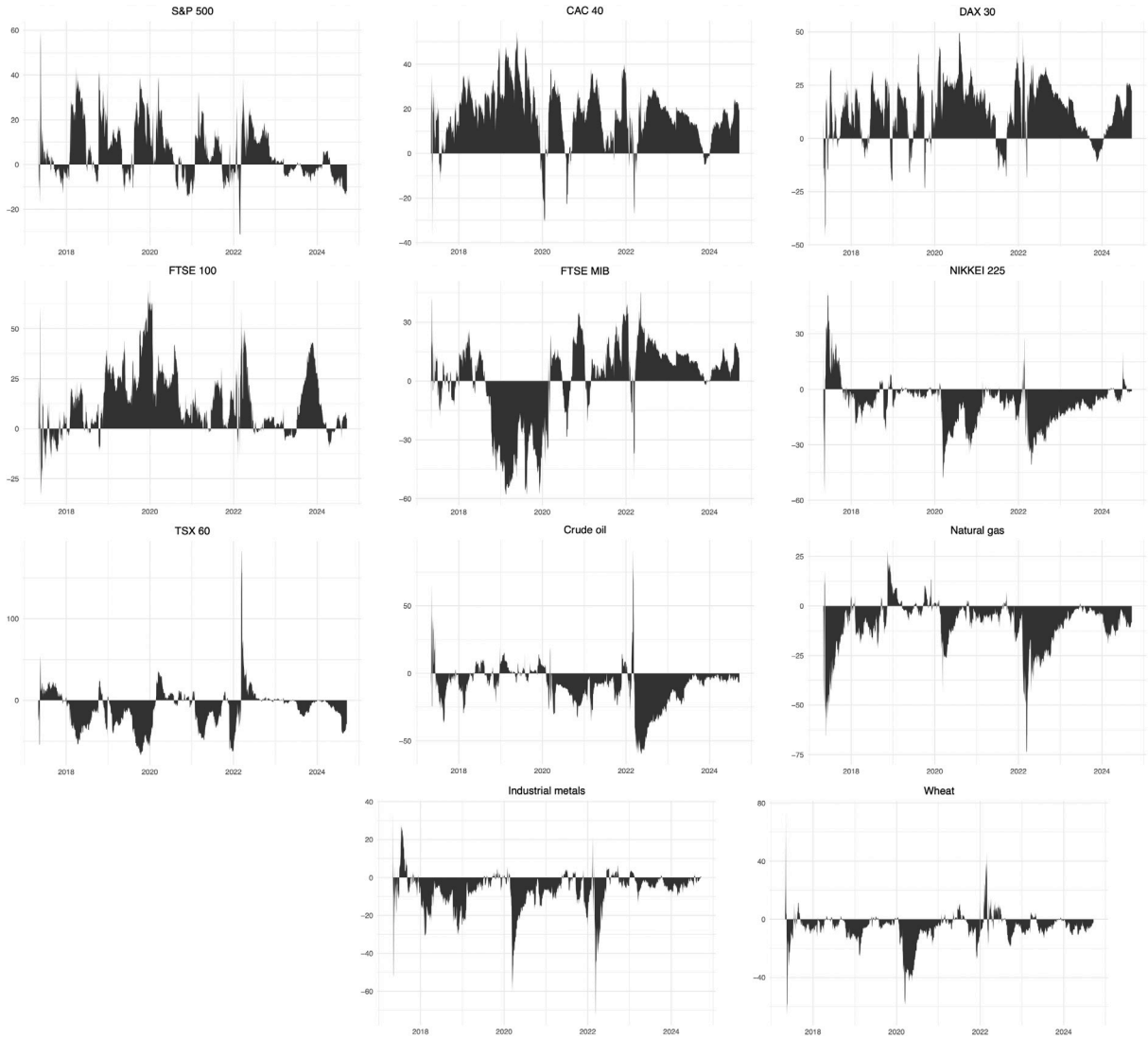

Fig. 3. Dynamic net directional return connectedness. Results are based on TVP-VAR model with lag length of BIC and a 10-step-ahead forecast.

Table 2
Averaged dynamic connectedness table.

	S&P500	CAC40	DAX30	FTSE100	FTSEMIB	NIKKEI225	TSX60	Crude oil	Natural gas	Metals	Wheat	FROM
Pre-conflict period												
S&P500	51.53	9.97	7.14	7.47	4.69	1.28	10.96	4.67	0.72	1.02	0.56	48.47
CAC40	8.48	34.52	16.96	15.00	14.38	1.56	5.03	1.66	0.75	1.11	0.56	65.48
DAX30	6.33	20.05	38.65	13.86	10.94	1.20	4.88	1.32	0.96	1.17	0.63	61.35
FTSE100	7.77	15.29	10.83	49.14	7.11	1.01	4.98	1.66	0.78	0.90	0.53	50.86
FTSEMIB	5.85	18.67	16.64	9.93	36.98	1.99	5.06	1.66	0.94	1.76	0.53	63.02
NIKKEI225	2.38	3.17	2.52	2.75	2.56	76.86	3.03	1.48	2.10	2.29	0.85	23.14
TSX60	16.11	9.39	7.56	8.42	5.41	1.63	41.75	7.15	0.87	1.13	0.57	58.25
Crude oil	6.55	3.81	2.69	2.90	2.66	1.97	6.45	68.81	1.82	1.16	1.18	31.19
Natural gas	1.89	1.30	1.58	1.15	1.12	2.10	2.07	3.18	83.56	1.13	0.91	16.44
Industrial metals	2.55	2.27	1.96	1.66	1.99	2.16	2.60	2.53	1.48	79.72	1.08	20.28
Wheat	1.06	1.59	1.55	1.13	1.26	0.90	0.81	1.15	1.13	0.72	88.70	11.30
TO	58.97	85.51	69.43	64.28	52.13	15.80	45.85	26.47	11.54	12.39	7.40	449.79
NET	10.51	20.03	8.08	13.42	−10.90	−7.34	−12.39	−4.72	−4.90	−7.89	−3.90	44.98/40.89
Conflict period												
S&P500	39.81	9.70	10.63	6.69	9.17	1.79	16.04	1.41	1.53	0.94	2.28	60.19
CAC40	7.48	28.33	21.88	9.78	19.97	1.65	6.34	1.08	1.00	0.94	1.54	71.67
DAX30	8.28	22.48	29.40	9.05	18.30	1.61	6.15	1.03	1.01	1.03	1.65	70.60
FTSE100	6.97	10.00	9.80	49.44	9.03	1.86	7.07	1.16	1.23	1.56	1.88	50.56
FTSEMIB	7.02	21.62	19.12	8.61	30.88	2.03	6.12	0.96	1.18	0.93	1.54	69.12
NIKKEI225	2.96	3.58	3.83	4.11	4.67	67.09	3.94	3.05	1.22	1.95	3.59	32.91
TSX60	16.36	8.68	8.09	9.17	8.47	2.44	39.13	2.67	1.60	1.47	1.92	60.87
Crude oil	3.84	2.91	2.69	3.72	2.79	3.10	5.59	65.74	2.07	2.31	5.24	34.26
Natural gas	3.77	2.98	3.43	2.88	3.33	2.46	2.83	1.56	71.10	3.47	2.19	28.90
Industrial metals	1.63	1.62	2.14	1.51	1.82	2.11	2.21	3.59	3.42	78.17	1.79	21.83
Wheat	3.61	2.39	2.76	4.62	2.52	2.85	3.15	1.32	1.44	2.14	73.19	26.81
TO	61.91	85.97	84.38	60.13	80.07	21.91	59.44	17.82	15.72	16.76	23.62	527.72
NET	1.72	14.30	13.79	9.57	10.95	−11.00	−1.43	−16.4	−13.19	−5.08	−3.18	52.77/47.97
All sample												
S&P500	50.28	9.95	8.16	6.51	6.34	0.95	12.27	3.44	0.82	0.77	0.52	49.72
CAC40	7.99	33.63	19.28	12.71	17.06	1.11	5.09	1.25	0.60	0.82	0.47	66.37
DAX30	6.76	21.52	36.74	12.07	14.08	0.87	4.83	0.99	0.72	0.87	0.54	63.26
FTSE100	6.90	13.04	9.95	53.54	7.52	0.96	4.86	1.34	0.69	0.75	0.46	46.46
FTSEMIB	6.16	20.36	17.98	9.10	36.10	1.44	5.10	1.31	0.72	1.26	0.48	63.90
NIKKEI225	1.72	2.36	1.88	2.14	2.16	81.51	2.20	1.78	1.62	1.89	0.74	18.49
TSX60	16.45	9.21	7.41	7.69	6.58	1.24	43.49	5.54	0.94	0.92	0.54	56.51
Crude oil	4.97	2.98	2.09	2.31	2.29	1.77	5.72	73.95	1.85	1.21	0.87	26.05
Natural gas	1.97	1.29	1.42	1.11	1.20	1.65	1.82	2.76	85.02	1.04	0.72	14.98
Industrial metals	1.88	1.76	1.60	1.39	1.60	1.57	2.07	2.25	1.37	83.61	0.89	16.39
Wheat	0.87	1.36	1.38	0.96	1.19	0.76	0.80	0.86	0.83	0.66	90.34	9.66
TO	55.67	83.83	71.14	55.97	60.01	12.31	44.76	21.51	10.16	10.18	6.24	431.79
NET	5.95	17.46	7.88	9.51	−3.89	−6.18	−11.75	−4.54	−4.82	−6.21	−3.42	43.18/39.25

Notes: Table presents the average dynamic connectedness among variables and total connectedness index (TCI), total directional connectedness TO others, total directional connectedness FROM others and the NET total directional connectedness.
The sample period (2 May 2017–17 September 2024) consists of 1834 observations. The Pre-conflict period (2 May 2017–23 February 2022) consists of 1179 observations. The Conflict period (24 February 2022–17 September 2024) consists of 655 observations.
Results are based on a TVP-VAR model with lag length of order 1 (BIC) in pre conflict period, order 2 (BIC) in conflict period, order 1 (BIC) for all sample and 10-step-ahead forecast.

In Fig. 3 is represented the dynamic connectedness of the net directional returns among markets. A positive value indicates that the information spillover is positive and, thus the related market behaves as a net transmitter. On the other hand, a negative value signals a negative information spillover which indicates that the market is a net receiver. The stock markets with more positive than negative values (with the exceptions of FTSE MIB although in part, NIKKEI 225 and TSX 60) are the main transmitters of information to other markets. While the commodity markets receive information from other markets for the entire period analyzed, highlighting, in this way, their distinct vulnerability. Furthermore, at the beginning of the conflict all markets show values that are elevated and in counter-trend in comparison to the previous and subsequent ones, even if this happens for a short time as in TSX 60 and natural gas market (Izzeldin et al. (2023)). An exception is the NIKKEI 225 which is less affected by the conflict, probably because it is distant and less dependent on the two belligerents. All commodity markets in February 2024, at the beginning of the conflict, are the main sources of shocks while stock markets seem to have a limited role in the system. After the conflict commodity markets, mainly crude oil and natural gas, become great receivers, i.e. they are dominated by other markets. From these considerations it emerges that unexpected and significant events, such as conflict, can modify at first and amplify later the performance connectedness among numerous stock and commodity markets (Benlagha and El Omari (2022) and Yuan et al. (2021)).

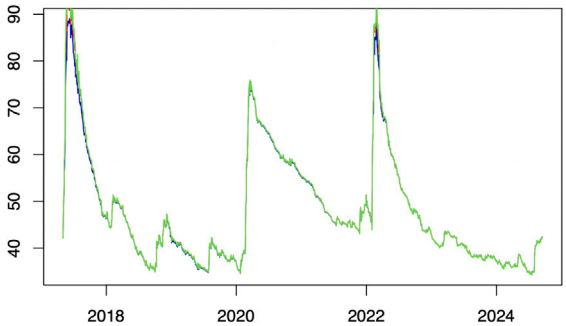

Fig. 4. Dynamic total return spillover using different forecast horizons.
Note: forecast horizons 5, 10, 20 are represented by red line, blue line and green line, respectively.

To test the stability of our results, the TVP-VAR estimate were performed using alternative forecast horizons, Fig. 4. The different line tends to overlap and are not completely remarkable. This consistency supports the robustness of our results regarding interconnectedness and spillovers among markets.

4. Conclusion

The Russia–Ukraine conflict has increased uncertainty and risk in commodity and financial markets. Consequently, it has altered economic dynamics and global balances. If the consequences of the war in the long term are uncertain, in the short term the prices of commodities have recorded considerable increases. The TVP-VAR model reveals financial risk coordination in the relationship between stock returns of G7 countries and commodities. Therefore, spillover transmission between analyzed markets respond differently at various event. Results distinguishing between net transmitters and net receivers within the network and in response to different events could help investors interested in the analyzed markets to build more efficient portfolios. In fact, given the temporal variability between commodities and stock markets and the long-lasting nature of the propagation of volatility spillover in the war period the portfolio weights should be adjusted. This is consistent with what Chancharat and Sinlapates (2023) shown regarding the correlation between oil and the Asian stock markets. Indeed, stock market present numerous risk and opportunities for investors particularly during times crisis, and this is what underlies changes in diversification strategies that must be adapted to the type of outbreak. This is because crisis periods showed more intense connectedness than normal periods. Furthermore, the results show how constraint and lack of diversification in the supply of raw materials is a risky factor for investors, but even more so for policymaker decisions. It is useful for policy makers to be able to distinguish the sources from which shocks originate and since information spillovers interact across different markets to adapt economic policy accordingly.

CRediT authorship contribution statement

Maria Leone: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Alberto Manelli:** Writing – review & editing, Supervision, Funding acquisition. **Roberta Pace:** Writing – review & editing, Writing – original draft, Supervision.

Acknowledgments

We would like to express our sincere gratitude to the editor, guest editors and the anonymous referees for their useful comments and suggestions.

Data availability

Data will be made available on request.

References

Adekoya, O.B., Oliyide, J.A., Noman, A., 2021. The volatility connectedness of the EU carbon market with commodity and financial markets in time- and frequency-domain: The role of the U.S. economic policy uncertainty. Resour. Policy 74, http://dx.doi.org/10.1016/j.resourpol.2021.102252.

Benlagha, N., El Omari, S., 2022. Connectedness of stock markets with gold and oil: new evidence from Covid-19 pandemic. Financ. Res. Lett. 46, http://dx.doi.org/10.1016/j.frl.2021.102373.

Burns, C.B., Prager, D.L., 2024. Do agricultural swaps co-move whith equity markets? Evidence from the COVID-19 crisis. J. Commod. Mark. 34, http://dx.doi.org/10.1016/j.jcomm.2024.100405.

Chancharat, S., Sinlapates, P., 2023. Dependencies and dynamic spillovers across crude oil and stock markets throughout the Covid-19 pandemic and Russia-Ukraine conflict: evidence from the ASEAN+6. Financ. Res. Lett. 57, http://dx.doi.org/10.1016/j.frl.2023.104249.

Chatziantoniou, I., Gabauer, D., 2021. EMU risk-synchronisation and financial fragility through the prism of dynamic connectedness. Q. Rev. Econ. Financ. 79, http://dx.doi.org/10.1016/j.qref.2020.12.003.

Cheng, Z., Li, M., Cui, R., Wei, Y., Wang, S., Hong, Y., 2024. The impact of COVID-19 on global financial markets: a multiscale volatility spillover analysis. Int. Rev. Financ. Anal. 95, http://dx.doi.org/10.1016/j.irfa.2024.103454.

Clancey-Shang, D., Fu, C., 2023. The Russia-Ukraine conflict and foreign stocks on the US market. J. Risk Financ. 24, http://dx.doi.org/10.1108/JRF-07-2022-0179.

Cui, J., Maghyereh, A., Liao, D., 2024. Risk connectedness between international oil and stock markets during the Covid-19 pandemic and the Russia-Ukraine conflict: fresh evidence from the higher-order moments. Int. Rev. Econ. Financ. 95, http://dx.doi.org/10.1016/j.iref.2024.103470.

Diebold, F.X., Yilmaz, K., 2014. On the network topology of variance decompositions: measuring the connectedness of financial firms. J. Econometrics 182, http://dx.doi.org/10.1016/j.jeconom.2014.04.012.

Dragomirescu-Gaina, C., Monticini, A., Salsano, F., 2024. Media news and market expectations: insights into the ECB's news data-dependent policy regime. Econom. Lett. 244, http://dx.doi.org/10.1016/j.econlet.2024.111987.

Elliott, G., Rothenberg, T.J., Stock, J.H., 1996. Efficient test for an autoregressive unit root. Econometrica 64, http://dx.doi.org/10.2307/2171846.

Gaio, L.E., Stefanelli, N.O., Tobajara, P.J., Bonacim, C.A.G., Gatsios, R.C., 2022. The impact of the Russia-Ukraine conflict on market efficiency: evidence for the developed stock market. Financ. Res. Lett. 50, http://dx.doi.org/10.1016/j.frl.2022.103302.

Inacio, C.M.C., Khristoufek, L., David, S.A., 2023. Assessing the impact of the Russia-Ukraine war on energy prices: a dynamic cross-correlation analysis. Phys. A: Stat. Math. Appl. 626, http://dx.doi.org/10.1016/j.physa.2023.129084.

Izzeldin, M., Muradoglu, Y.G., Pappas, V., Petropoulou, A., Sivaprasad, S., 2023. The impact of the Russia-Ukrainian war on global financial markets. Int. Rev. Financ. Anal. 87, http://dx.doi.org/10.1016/j.irfa.2023.102598.

Koop, G., Korobilis, D., 2014. A new index of financial conditions. Eur. Econ. Rev. 71, http://dx.doi.org/10.1016/j.euroecorev.2014.07.002.

Koop, G., Pesaran, M., Potter, S.M., 1996. Impulse response analysis in nonlinear multivariate models. J. Econometrics 74, http://dx.doi.org/10.1016/0304-4076(95)01753-4.

Laurent, S., Shi, S., 2022. Unit root test with high-frequency data. Econometric Theory 38, http://dx.doi.org/10.1017/S0266466621000098.

Lee, G., Silva, N., Jeong, J., 2024. The effect of Russia-Ukraine war on stock market linkages beyond its borders: evidence from Latin America. Appl. Econ. Lett. 1–12. http://dx.doi.org/10.1080/13504851.2024.2364015.

Lei, L., Aziz, G., Sarwar, S., Waheed, R., Tiwari, A.K., 2023. Spillover and portfolio analysis for oil and stock market: a new insight across financial crisis, COVID-19 and Russian-Ukraine war. Resour. Policy 85, http://dx.doi.org/10.1016/j.resourpol.2023.103645.

Lin, Z.L., Ouyang, W.P., Yu, Q.R., 2024. Risk spillover effects of the Israel-Hamas War on global financial and commodity markets: a time-frequency and networks analysis. Financ. Res. Lett. 66, http://dx.doi.org/10.1016/j.frl.2024.105618.

Ma, D., Zhang, Y., Ji, Q., Zhao, W.L., Zhai, P., 2024. Heterogeneous impact of climate change news on China's financial markets. Int. Rev. Financ. Anal. 91, http://dx.doi.org/10.1016/j.irfa.2023.103007.

Pan, Q., Sun, Y., 2023. Changes in volatility leverage and spillover effects of crude oil futures markets affected by the 2022 Russia-Ukraine conflict. Financ. Res. Lett. 58, http://dx.doi.org/10.1016/j.frl.2023.104442.

Umar, Z., Polat, O., Choi, S.Y., Teplova, T., 2022. The impact of the Russia-Ukraine conflict on the connectedness of financial markets. Financ. Res. Lett. 48, http://dx.doi.org/10.1016/j.frl.2022.102976.

Wu, Y., Ren, W., Wan, J., Liu, X., 2023. Time-frequency volatility connectedness between fossil energy and agricultural commodities: comparing the COVID-19 pandemic with the Russia-Ukraine conflict. Financ. Res. Lett. 55, http://dx.doi.org/10.1016/j.frl.2023.103866.

Yousaf, I., Hunjra, A.I., Alshater, M.M., Bouri, E., Li, I., 2023. Multidimensional connectedness among the volatility of global financial markets around the Russian-Ukrainian conflict. Pac.-Basin Financ. J. 82, http://dx.doi.org/10.1016/j.pacfin.2023.102163.

Yuan, D., Zhang, F., Cui, F., Wang, S., 2021. Oil and BRIC stock markets before and after COVID-19: a local gaussian correlation approach. Emerg. Mark. Financ. Trade 57, http://dx.doi.org/10.1080/1540496X.2021.1904886.

Zhang, Y., Sun, Y., 2023. Did U.S. and Chinese investors respond differently to the exogenous shocks from Covid-19 and the war in Ukraine? Int. Rev. Financ. Anal. 88, http://dx.doi.org/10.1016/j.irfa.2023.102710.